Lecture Notes on Hyperbolic PDEs

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Preface

Lecture 1: Introduction (2026/4/24)

1.1 Main topics

In this lecture we mainly talk about the theory of first-order systems of hyperbolic conservation laws, which are widely applied in gas dynamics, combustion theory (detonation waves), plasma physics (astrophysics), water wave theory, and so on.

1.2 Definitions

The conservation laws can be described as follows

$$u_t + f(u)_x = 0, \tag{1}$$

where $u = (u_1, \cdots, u_n)^\top \in \mathbb{R}^n$, $n \geq 1$, $u_i = u_i(x,t)$, $(x,t) \in \mathbb{R} \times \mathbb{R}_+$, $i = 1, \cdots, n$, and $f(u) = (f_1(u), \cdots, f_n(u))^\top \in C^2(\Omega)$, $\Omega \subset \mathbb{R}^n$.

We can write the above conservation laws (1) in matrix form

$$\begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}_t + \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_n} \end{pmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}_x = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \tag{2}$$

or abbreviated as

$$u_t + A(u)u_x = 0, \tag{3}$$

where the matrix function

$$A(u) = \begin{pmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_n} \end{pmatrix}.$$

We say conservation laws (1) is *hyperbolic* if $A(u)$ has n linearly independent right eigenvectors; and is *strictly hyperbolic* if $A(u)$ has n distinct real eigenvalues $\lambda_1(u) < \cdots < \lambda_n(u)$.

1.3 Examples

Let us consider some examples of conservation laws and determine the type of them.

Example 1.1 (p-system)**.**

$$\begin{cases} \tau_t - u_x = 0, \\ u_t + p_x = 0, \end{cases} \tag{4}$$

where τ stands for the specific volume (the reciprocal of the mass density ρ), i.e., $\tau = 1/\rho$, u represents the fluid velocity, and p denotes the pressure and is the function of τ defined as $p(\tau) = \tau^{-\gamma}$ with $1 < \gamma < 3$. γ is polytropic index or adiabatic index.

We can write (4) in matrix form

$$\begin{pmatrix} \tau \\ u \end{pmatrix}_t + \begin{pmatrix} 0 & -1 \\ p'(\tau) & 0 \end{pmatrix} \begin{pmatrix} \tau \\ u \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and compute the eigenvalues by sovling the following characteristic equation

$$\begin{vmatrix} \lambda & 1 \\ -p'(\tau) & \lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + p'(\tau) = 0 \Rightarrow \lambda = \pm\sqrt{-p'(\tau)}.$$

The expression of eigenvalues is valid since $p'(\tau) = -\gamma\tau^{-\gamma-1} < 0$. Then the conservation laws (4) has two distinct eigenvalues $\lambda_1 = -\sqrt{-p'(\tau)}$ and $\lambda_2 = \sqrt{-p'(\tau)}$, hence p-system is strictly hyperbolic.

Example 1.2 (nonlinear wave system)**.**

$$\begin{cases} \rho_t + m_x = 0, \\ m_t + p_x = 0, \end{cases} \tag{5}$$

where $m = \rho u$ denotes the momentum density, i.e., momentum per unit volume, and here p is the function of ρ defined as $p = \rho^\gamma$ with $1 < \gamma < 3$.

We can write (5) in matrix form

$$\begin{pmatrix} \rho \\ m \end{pmatrix}_t + \begin{pmatrix} 0 & 1 \\ p'(\rho) & 0 \end{pmatrix} \begin{pmatrix} \rho \\ m \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence we get the eigenvalues from the characteristic function

$$\begin{vmatrix} \lambda & 1 \\ p'(\rho) & \lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - p'(\rho) = 0 \Rightarrow \lambda = \pm\sqrt{p'(\rho)}.$$

The expression above is valid since $p'(\rho) = \gamma\rho^{\gamma-1} > 0$. Then nonlinear wave system has two distinct eigenvalues $\lambda_1 = -\sqrt{p'(\rho)}$ and $\lambda_2 = \sqrt{p'(\rho)}$, and hence the system is strictly hyperbolic. We call (5) wave system for the reason that we can take the partial derivative with respect to t in the first equation and take the partial derivative with respect to x in the second equation, then we get

$$\begin{cases} \rho_{tt} + m_{xt} = 0, \\ m_{tx} + p_{xx} = 0. \end{cases}$$

Hence $\rho_{tt} - (p(\rho))_{xx} = 0$ by eliminating the same part $m_{xt} = m_{tx}$. This is a nonlinear wave equation.

For an equation like (1), we need to provide appropriate auxiliary conditions (initial and/or boundary conditions) so that the problem is well-posed (i.e., a unique solution exists and depends continuously on the given data).

Now we consider the following Cauchy problem

$$\begin{cases} u_t + f(u)_x = 0, & (x,t) \in \mathbb{R} \times \mathbb{R}_+, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases} \tag{6}$$

Our aim is to find the classical solution on $D \subset \mathbb{R} \times \mathbb{R}_+$ to problem (6).

We first give an example of simple case.

Example 1.3 (Burgers' equation)**.** Given $n = 1$, and consider

$$\begin{cases} u_t + \left(\dfrac{u^2}{2}\right)_x = 0, & \text{(7a)} \\ u(x,0) = u_0(x) = x. & \text{(7b)} \end{cases}$$

Next we prove that the problem (7) admits global classical solution in time.

Proof. *For any fixed point $\xi \in \mathbb{R}$, define the characteristic line through $(\xi, 0)$ (see the blue line in Figure 1) of (7a) as follows*

$$x - \xi = \xi(t - 0) \Rightarrow x = \xi + \xi t. \tag{8}$$

Then we can solve

$$\xi = \frac{x}{1+t}.$$

This shows that the slope of characteristic line can be uniquely determined by $(x,t) \in \mathbb{R} \times \mathbb{R}_+$. Now we write (7a) in the form

$$u_t + uu_x = 0.$$

Then u is a constant along the characteristic line (8), since the left-hand side represents the directional derivative of u along this characteristic line. Hence $u = u(\xi, 0) = u_0(\xi) = \xi = \frac{x}{1+t}$ for $t > 0$. □

Remark 1.1. In fact, we can fix any point $(x_0, t_0) \in \mathbb{R} \times \mathbb{R}_+$ instead of those just on x axis. The slope of the characteristic line is $\frac{x_0}{1+t_0}$ under this circumstance, and the characteristic line (see the red line of Figure 1) is

$$x - x_0 = \frac{x_0}{1+t_0}(t - t_0).$$

Moreover, the monotonicity of initial condition $u_0(x)$ with respect to variable x is important. If $u_0(x)$ is a *nondecreasing* function of x, responding to the situation that all characteristic lines diverge, the problem (7) admits unique global classical solution in time. More general cases have been discussed in [1].

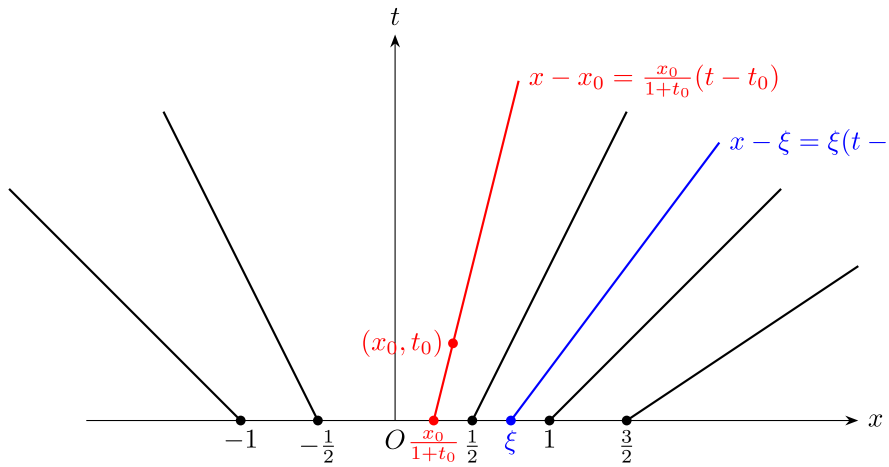

图 1: Characteristic lines of scalar conservation laws

Now we discuss the solution to the problem (7) replacing the initial condition (7b) with

$$u(x,0) = \begin{cases} 1, & x < 0, \\ 1 - x, & 0 < x < 1, \\ 0, & x > 1. \end{cases} \tag{9}$$

As is shown in Figure 2, u is a piecewise function of x.

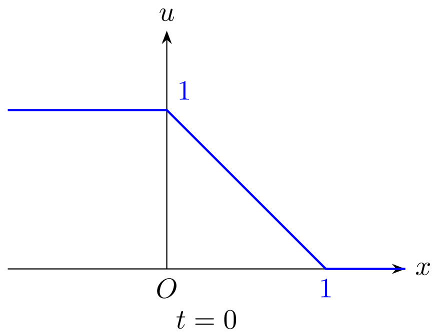

图 2: The initial function is piecewise

We notice that the slopes of characteristic lines are always 1 and 0 when $x < 0$ and $x > 1$ respectively. Now given any fixed point $(x_0, t_0) \in \mathbb{R} \times \mathbb{R}_+$ with $0 < x < 1$, we can determine the slope of characteristic lines, namely $\frac{1-x_0}{1-t_0}$, and characteristic lines can be written in

$$x - x_0 = \frac{1 - x_0}{1 - t_0}(t - t_0), \quad 0 < x < 1.$$

Hence the classical solution (see Figure 3) to Cauchy problem (7a)–(9) is

$$u(x,t) = \begin{cases} 1, & x < t, \\ \dfrac{1-x}{1-t}, & t < x < 1, \\ 0, & x > 1, \end{cases}$$

where $0 < t < 1$. This shows that the solution only exists in finite time. We can also draw some special cases to understand, see Figure 4.

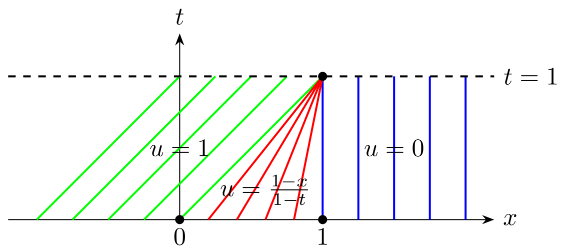

图 3: Solution to scalar conservation laws with a piecewise initial condition

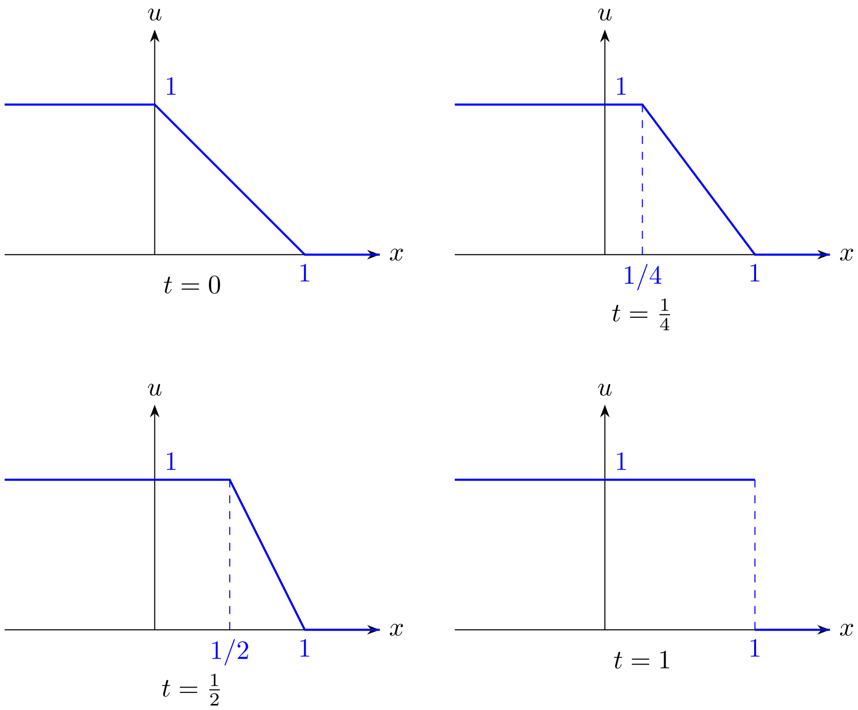

图 4: More special cases

Therefore we must generalize the notion of solution from classical solution to weak solution or distributional solution, which was proposed by Riemann in 1860 [2].

1.4 Generalized Solutions

Next we consider the general Cauchy problem for conservation laws

$$\begin{cases} u_t + f(u)_x = 0, \quad (x,t) \in \mathbb{R} \times \mathbb{R}_+, & \text{(10a)} \\ t = 0: \quad u(x,0) = \varphi(x), \quad x \in \mathbb{R}. & \text{(10b)} \end{cases}$$

Here $u(x,t)$ is just a bounded and measurable function. Let $\phi \in C_0^1(\mathbb{R}\times[0,\infty))$, say u is a *generalized solution* or *distribution* to (10), if

$$\int_{\mathbb{R}} \int_{t>0} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} \varphi(x)\phi(x,0)\mathrm{d}x = 0, \tag{11}$$

where ϕ is also called *test functions*.

Lecture 2: (2026/5/8)

2.1 The Relation Between Classical Solutions and Generalized Solutions

We can prove if $u = u(x,t)$ is the classical solution to Cauchy problem

$$\begin{cases} u_t + f(u)_x = 0, & (x,t) \in \mathbb{R} \times \mathbb{R}_+, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases} \tag{12}$$

Then u is the generalized solution, namely

$$\int_{\mathbb{R}} \int_{t>0} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} u_0(x)\phi(x,0)\mathrm{d}x = 0 \tag{13}$$

holds for all $\phi \in C_0^\infty(\mathbb{R} \times [0,\infty))$.

Proof. *Suppose u is the classical solution to problem (12). Let $[a,b] \times [0,T]$ be the rectangular domain that contains $\operatorname{supp}\phi$ (see Figure 5). Then*

$$\begin{aligned}
&\int_{\mathbb{R}} \int_{t>0} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} u_0(x)\phi(x,0)\mathrm{d}x \\
=& \int_a^b \int_0^T (u\phi_t + f(u)\phi_x)\mathrm{d}t\mathrm{d}x + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x \\
=& \int_a^b \int_0^T [(u\phi)_t + (f(u)\phi)_x - \phi(u_t + f(u)_x)]\mathrm{d}t\mathrm{d}x + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x \\
=& \int_a^b \int_0^T [(u\phi)_t + (f(u)\phi)_x]\mathrm{d}t\mathrm{d}x + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x \\
=& \int_a^b \int_0^T (u\phi)_t\mathrm{d}t\mathrm{d}x + \int_0^T \int_a^b (f(u)\phi)_x\mathrm{d}x\mathrm{d}t + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x \\
=& \int_a^b [u(x,T)\phi(x,T) - u(x,0)\phi(x,0)]\mathrm{d}x + \int_0^T [f(u(b,t))\phi(b,t) - f(u(a,t))\phi(a,t)]\mathrm{d}t + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x \\
=& -\int_a^b u_0(x)\phi(x,0)\mathrm{d}x + \int_a^b u_0(x)\phi(x,0)\mathrm{d}x = 0.
\end{aligned}$$

□

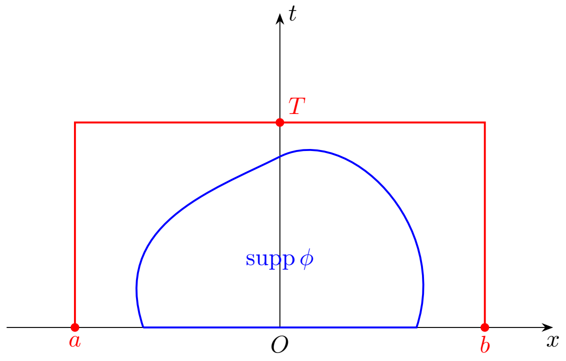

图 5: The rectangular domain

2.2 Rankine-Hugoniot Relation

We next show that not every discontinuity is permissible, provided that the condition (13) places severe restrictions on the curves of discontinuity, which will be later introduced as *Rankine-Hugoniot relation*.

Let Γ be a smooth curve on (x,t) plane, and across which u has a jump discontinuity (i.e., u has well defined limits on both sides of Γ, and u is smooth away from Γ). Let P be any point on Γ and D be a small ball centered at P. Assume that in D, Γ is given by $x = x(t)$. Let D_1 and D_2 be the components of D which are determined by Γ (see Figure 6).

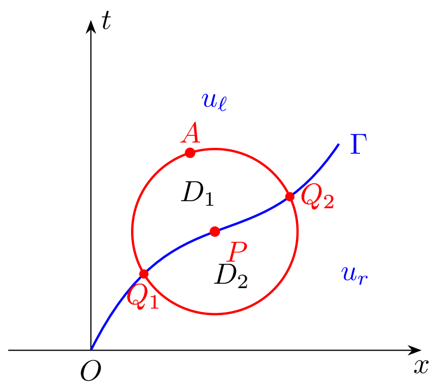

图 6: The discontinuity curve

Choose $\phi \in C_0^1(D)$, then

$$
\begin{aligned}
0 &= \int_{\mathbb{R}} \int_{t>0} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t + \int_{\mathbb{R}} u_0(x)\phi(x,0)\mathrm{d}x \\
&= \int_D (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t \\
&= \int_{D_1} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t + \int_{D_2} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t.
\end{aligned}
$$

Notice that

$$
\begin{aligned}
\int_{D_i} (u\phi_t + f(u)\phi_x)\mathrm{d}x\mathrm{d}t &= \int_{D_i} [(u\phi)_t + (f(u)\phi)_x - \phi(u_t + f(u)_x)]\mathrm{d}x\mathrm{d}t \\
&= \int_{D_i} [(u\phi)_t + (f(u)\phi)_x]\mathrm{d}x\mathrm{d}t \\
&= \int_{D_i} \nabla \cdot (f(u)\phi, u\phi)\mathrm{d}x\mathrm{d}t \quad \text{(divergence theorem)} \\
&= \oint_{\partial D_i} (f(u)\phi, u\phi) \cdot \nu\mathrm{d}s \quad (\nu \text{ is the unit outward normal vector of } \partial D_i) \\
&= \oint_{\partial D_i} (f(u)\phi, u\phi) \cdot (\mathrm{d}t, -\mathrm{d}x) \quad (\nu\mathrm{d}s = (\mathrm{d}t, -\mathrm{d}x)) \\
&= \oint_{\partial D_i} \phi(-u\mathrm{d}x + f(u)\mathrm{d}t).
\end{aligned}
$$

Since $\phi = 0$ on ∂D, then

$$\begin{aligned}\oint_{\partial D_1} \phi(-u\mathrm{d}x + f(u)\mathrm{d}t) &= \int_{\widetilde{Q_1Q_2}+\widetilde{Q_2AQ_1}} \phi(-u\mathrm{d}x + f(u)\mathrm{d}t)\\ &= \int_{\widetilde{Q_1Q_2}} \phi(-u\mathrm{d}x + f(u)\mathrm{d}t)\\ &= \int_{\widetilde{Q_1Q_2}} \phi(-u_\ell\mathrm{d}x + f(u_\ell)\mathrm{d}t).\end{aligned}$$

and

$$\oint_{\partial D_2} \phi(-u\mathrm{d}x + f(u)\mathrm{d}t) = \int_{\widetilde{Q_2Q_1}} \phi(-u_r\mathrm{d}x + f(u_r)\mathrm{d}t) = -\int_{\widetilde{Q_1Q_2}} \phi(-u_r\mathrm{d}x + f(u_r)\mathrm{d}t).$$

Hence

$$\int_{\widetilde{Q_1Q_2}} \phi(-u_\ell\mathrm{d}x + f(u_\ell)\mathrm{d}t) - \int_{\widetilde{Q_1Q_2}} \phi(-u_r\mathrm{d}x + f(u_r)\mathrm{d}t) = 0.$$

This implies

$$\int_{\widetilde{Q_1Q_2}} \phi(-(u_\ell - u_r)\mathrm{d}x + (f(u_\ell) - f(u_r))\mathrm{d}t) = 0.$$

Let $[u] := u_\ell - u_r$ and $[f(u)] := f(u_\ell) - f(u_r)$, then

$$\int_{\widetilde{Q_1Q_2}} \phi(-[u]\mathrm{d}x + [f(u)]\mathrm{d}t) = 0.$$

Since ϕ is arbitrary, we conclude that

$$s[u] = [f(u)] \tag{14}$$

on Γ, where $s = \frac{\mathrm{d}x}{\mathrm{d}t}$ denotes the propagation speed of the discontinuity. Relation (14) is known as *Rankine-Hugoniot relation.*

2.3 Construct Piecewise Smooth Solutions

Now we can utilize the relation (14) to construct piecewise smooth solutions.

Example 2.1 (Burgers' equation)**.** We still give an example of scalar conservation laws

$$\begin{cases} u_t + \left(\dfrac{u^2}{2}\right)_x = 0, & (x,t) \in \mathbb{R} \times \mathbb{R}_+,\\ u(x,0) = \begin{cases} 1, & x < 0,\\ 1 - x, & 0 < x < 1,\\ 0, & x > 1. \end{cases} & x \in \mathbb{R}. \end{cases}$$

We have discussed such Cauchy problem has classical solution only in $\mathbb{R} \times (0,1)$. Now we are going to construct a generalized solution satisfying relation (14). When $t \geq 1$, we take $u_\ell = 1$ and $u_r = 0$, then

$$s = \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{[f(u)]}{[u]} = \frac{\frac{1^2}{2} - \frac{0^2}{2}}{1 - 0} = \frac{1}{2}.$$

Hence the generalized solution is

$$u(x,t) = \begin{cases} 1, & x < t,\ 0 < t < 1,\\ \dfrac{1-x}{1-t}, & t < x < 1,\ 0 < t < 1,\\ 0, & x > 1,\ 0 < t < 1,\\ 1, & x < \dfrac{t+1}{2},\ t > 1,\\ 0, & x > \dfrac{t+1}{2},\ t > 1. \end{cases}$$

As is shown in Figure 7.

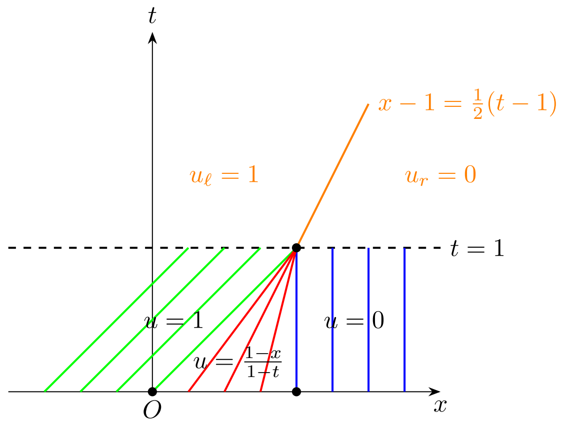

图 7: A generalized solution to scalar conservation laws

2.4 The Riemann Problem

Riemann proposed a special problem in [2] in 1860, which later called the Riemann problem. Now we consider this problem

$$\begin{cases} u_t+f(u)_x=0, & (x,t)\in\mathbb{R}\times\mathbb{R}_+, \\ u(x,0)=\begin{cases} u_\ell, & x>0, \\ u_r, & x<0. \end{cases} & x\in\mathbb{R}, \end{cases} \tag{15}$$

where $u_\ell, u_r\in\mathbb{R}^n$ are two constant vectors.

2.5 The self-similar solution and rarefaction wave

Lecture 3: (2026/5/9)

Lecture 4: (2026/5/11)

Problem 4.1. Consider the system of conservation laws

$$\begin{cases} \rho_t + m_x = 0, \\ m_t + p_x = 0. \end{cases} \tag{16}$$

with Riemann initial data

$$(\rho, m)(x, 0) = \begin{cases} (\rho_\ell, m_\ell), & x < 0, \\ (\rho_r, m_r), & x > 0. \end{cases} \tag{17}$$

Here $p(\rho) = \rho^\gamma$ with $1 < \gamma < 3$.

Solution. *Firstly we write the system in vector form*

$$U_t + A(U)U_x = 0,$$

where

$$U = \begin{pmatrix} \rho \\ m \end{pmatrix} \quad \text{and} \quad A(U) = \begin{pmatrix} 0 & 1 \\ p'(\rho) & 0 \end{pmatrix}.$$

The eigenvalues of $A(U)$ are

$$\lambda_1 = -\sqrt{p'(\rho)} \quad \text{and} \quad \lambda_2 = \sqrt{p'(\rho)}.$$

Next we will find shock solutions *and* rarefaction wave solutions *to this Riemann problem.*

1. *Shock solutions. We first consider that when right-hand condition $(\rho, m)^\top$ can be connected with a 1-shock with the given left-hand condition $(\rho_\ell, m_\ell)^\top$. Then by Rankine-Hugoniot condition and Lax entropy condition we get*

   $$s(U_\ell - U) = F(U_\ell) - F(U),$$

   where $F = (m, p)^\top$. That is to say

   $$s\begin{pmatrix} \rho_\ell - \rho \\ m_\ell - m \end{pmatrix} = \begin{pmatrix} m_\ell - m \\ p_\ell - p \end{pmatrix}$$

   and

   $$\lambda_1(U_\ell) > s > \lambda_1(U),$$

   where $p_\ell = p(\rho_\ell)$ and $U_\ell = (\rho_\ell, m_\ell)^\top$. Hence

   $$\frac{s(\rho_\ell - \rho)}{s(m_\ell - m)} = \frac{m_\ell - m}{p_\ell - p} \Rightarrow (m_\ell - m)^2 = (\rho_\ell - \rho)(p_\ell - p)$$
   $$\Rightarrow m_\ell - m = \pm\sqrt{(\rho_\ell - \rho)(p_\ell - p)}.$$

   Since

   $$-\sqrt{p'(\rho_\ell)} > s > -\sqrt{p'(\rho)} \Rightarrow \rho_\ell < \rho \quad \text{and} \quad s < 0,$$

   we get $p_\ell < p$. Notice

   $$s(\rho_\ell - \rho) = m_\ell - m > 0.$$

   Then

   $$m_\ell - m = \sqrt{(\rho_\ell - \rho)(p_\ell - p)} = \sqrt{(\rho_\ell - \rho)(\rho_\ell^\gamma - \rho^\gamma)}.$$

Hence the 1-shock solution is

$$u = \frac{\rho_\ell u_\ell - \sqrt{(\rho-\rho_\ell)(\rho^\gamma - \rho_\ell^\gamma)}}{\rho} \quad (\rho > \rho_\ell), \tag{18}$$

and the speed of 1-shock wave is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = s_1 = \frac{m_\ell - m}{\rho_\ell - \rho} = -\frac{m_\ell - m}{\rho - \rho_\ell} = -\frac{\rho_\ell u_\ell - \rho u}{\rho - \rho_\ell} < 0. \tag{19}$$

Next we find 2-shock solution. Since

$$\lambda_2(U_\ell) > s > \lambda_2(U),$$

we get

$$\sqrt{p'(\rho_\ell)} > s > \sqrt{p'(\rho)} \Rightarrow \rho_\ell > \rho \quad \textit{and} \quad s > 0.$$

Notice that

$$s(\rho_\ell - \rho) = m_\ell - m > 0.$$

Then

$$m_\ell - m = \sqrt{(\rho_\ell - \rho)(p_\ell - p)} = \sqrt{(\rho_\ell - \rho)(\rho_\ell^\gamma - \rho^\gamma)}.$$

Hence the 2-shock solution is

$$u = \frac{\rho_\ell u_\ell - \sqrt{(\rho_\ell-\rho)(\rho_\ell^\gamma - \rho^\gamma)}}{\rho} \quad (\rho_\ell > \rho), \tag{20}$$

and the speed of 2-shock wave is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = s_2 = \frac{m_\ell - m}{\rho_\ell - \rho} = \frac{\rho_\ell u_\ell - \rho u}{\rho_\ell - \rho} > 0. \tag{21}$$

2. *Rarefaction wave solutions. Now we will find the self-similar solution like $u = u(\xi)$ to this Riemann problem, where $\xi = x/t$. Since the system can be written as*

$$\begin{pmatrix}\rho\\ m\end{pmatrix}_t + \begin{pmatrix}0 & 1\\ p'(\rho) & 0\end{pmatrix}\begin{pmatrix}\rho\\ m\end{pmatrix}_x = \begin{pmatrix}0\\ 0\end{pmatrix}.$$

Then by chain rule we have

$$-\frac{x}{t^2}\begin{pmatrix}\rho\\ m\end{pmatrix}_\xi + \frac{1}{t}\begin{pmatrix}0 & 1\\ p'(\rho) & 0\end{pmatrix}\begin{pmatrix}\rho\\ m\end{pmatrix}_\xi = \begin{pmatrix}0\\ 0\end{pmatrix}.$$

That is

$$\begin{pmatrix}0 & 1\\ p'(\rho) & 0\end{pmatrix}\begin{pmatrix}\rho\\ m\end{pmatrix}_\xi = \xi\begin{pmatrix}\rho\\ m\end{pmatrix}_\xi.$$

Then ξ is an eigenvalue of matrix $A(u)$, and $(\rho, m)_\xi^\top$ is the corresponding eigenvetor. Hence if

$$\frac{x}{t} = \xi = -\sqrt{p'(\rho)},$$

then we get

$$m_\xi = \xi\rho_\xi \Rightarrow \frac{\mathrm{d}m}{\mathrm{d}\rho} = -\sqrt{p'(\rho)}.$$

Integrate from ρ_ℓ to ρ we have

$$m - m_\ell = \int_{\rho_\ell}^{\rho} -\sqrt{p'(y)}\mathrm{d}y.$$

Meanwhile we also need

$$\lambda_1(\rho_\ell) < \lambda_1(\rho) \Rightarrow -\sqrt{p'(\rho_\ell)} < -\sqrt{p'(\rho)} \Rightarrow \rho_\ell > \rho.$$

Hence we get 1-rarefaction wave solution

$$u = \frac{\rho_\ell u_\ell - \int_{\rho_\ell}^{\rho} \sqrt{p'(y)}\mathrm{d}y}{\rho} = \frac{\rho_\ell u_\ell - \frac{2\sqrt{\gamma}}{\gamma+1}\left(\rho^{\frac{\gamma+1}{2}} - \rho_\ell^{\frac{\gamma+1}{2}}\right)}{\rho} \quad (\rho < \rho_\ell). \tag{22}$$

Similarily, we have 2-rarefaction wave solution

$$u = \frac{\rho_\ell u_\ell + \left(\frac{2\sqrt{\gamma}}{\gamma+1}\rho^{\frac{\gamma+1}{2}} - \frac{2\sqrt{\gamma}}{\gamma+1}\rho_\ell^{\frac{\gamma+1}{2}}\right)}{\rho} \quad (\rho > \rho_\ell). \tag{23}$$

□

Lecture 5: 一维理想流体 Riemann 问题解的构造 (2026/5/15)

本节我们考虑如下一维理想流体 Riemann 问题解的构造:

$$
\begin{cases}
\begin{cases}
\tau_t - u_x = 0, \ (\text{质量守恒}) \\
u_t + p_x = 0, \ (\text{动量守恒}) \\
E_t + (up)_x = 0, \ (\text{能量守恒})
\end{cases} \\
(u,\tau,s)(x,0) = \begin{cases} (u_\ell,\tau_\ell,s_\ell), \ x<0, \\ (u_r,\tau_r,s_r), \ x>0. \end{cases}
\end{cases}
\tag{24}
$$

5.1 将能量守恒转化为熵守恒

Notation. $E = e + \frac{1}{2}u^2$ 表示单位质量的总能量 (内能 + 动能). 单位质量的内能 e 满足热力学基本关系式: $\mathrm{d}e = T\mathrm{d}s - p\mathrm{d}\tau$, 其中 T 为温度, s 为熵, p 为压强, τ 为比容且与密度 ρ 之间满足关系 $\tau = 1/\rho$. 由全微分的定义, 对函数 $e = e(s,\tau)$ 进行全微分并比较系数就有

$$
\begin{cases}
\dfrac{\partial e}{\partial s} = T, \\
\dfrac{\partial e}{\partial \tau} = -p.
\end{cases}
$$

在理想流体的假设下, 状态方程满足 $p\tau = RT$. 一类特殊的理想流体, 即多方气体, 满足 $e = C_vT$, 其中 C_v 为常数. 因为 $\tau = 1/\rho$, 于是 $\mathrm{d}\tau = -\mathrm{d}\rho/\rho^2$, 从而

$$
\mathrm{d}e = T\mathrm{d}s + p\frac{\mathrm{d}\rho}{\rho^2} = T\mathrm{d}s + RT\rho\frac{\mathrm{d}\rho}{\rho^2} \Rightarrow C_v\frac{\mathrm{d}e}{e} = \mathrm{d}s + R\frac{\mathrm{d}\rho}{\rho}.
$$

积分有

$$
\begin{aligned}
&C_v \log e = s + R\log\rho + \text{const}. \\
\Rightarrow & e = \text{const}.\ \exp\left\{\frac{1}{C_v}(s + R\log\rho)\right\} = \text{const}.\ \left(\exp\left\{\frac{s}{C_v}\right\}\rho^{R/C_v}\right).
\end{aligned}
$$

因此

$$
p = \frac{R}{C_v}e\rho = \frac{R}{C_v}\text{const}.\ \left(\exp\left\{\frac{s}{C_v}\right\}\rho^{R/C_v}\right)\rho = \text{const}.\ \exp\left\{\frac{s}{C_v}\right\}\rho^{1+R/C_v}.
$$

简记为 $\rho = A(s)\rho^\gamma$, 其中 $A(s) = \exp\{s/s_0\}$, $s_0 = C_v$ (方便起见可设为 1), $\gamma = 1 + R/C_v \in (1,3)$. 于是由 $p\tau = RT$ 可知

$$
e = C_vT = C_v\frac{1}{R}p\tau = \frac{p\tau}{R/C_v} = \frac{p\tau}{\gamma-1} = \frac{A(s)\rho^{\gamma-1}}{\gamma-1} = \frac{A(s)\tau^{1-\gamma}}{\gamma-1}.
$$

因此

$$
\begin{aligned}
&\frac{\partial e}{\partial s} = \frac{1}{s_0}\frac{A(s)\tau^{1-\gamma}}{\gamma-1} = \frac{e}{s_0}. \\
&\frac{\partial e}{\partial \tau} = -A(s)\tau^{-\gamma} = -A(s)\rho^\gamma = -p. \\
&p\tau = (\gamma-1)e = \underbrace{(\gamma-1)s_0}_{=R}T.
\end{aligned}
$$

从 $e = e(s,\tau)$ 出发我们有

$$
e_t = e_s s_t + e_\tau \tau_t =
$$

Lecture 6: 非线性波方程 Riemann 问题解的构造以及一般的一维守恒律 Riemann 问题的求解 (2026/5/18)

6.1 非线性波方程 Riemann 问题解的构造

我们现在考虑下面这个 Riemann 问题:

$$\begin{cases} \begin{cases} \rho_t + m_x = 0, \\ m_t + p_x = 0. \end{cases} \\ (\rho, m)(x, 0) = \begin{cases} (\rho_\ell, m_\ell), & x < 0, \\ (\rho_r, m_r), & x > 0. \end{cases} \end{cases} \tag{25}$$

6.2 一般的一维守恒律 Riemann 问题的求解

Lecture 7: (2026/5/22)

一道作业题: 求解如下一维等熵可压缩欧拉方程组的 Riemann 问题:

$$
\begin{cases}
\begin{cases}
\rho_t + (\rho u)_x = 0, \\
(\rho u)_t + (\rho u^2 + p)_x = 0,
\end{cases} \\
(\rho, u)(x, 0) = \begin{cases}
(\rho_\ell, u_\ell), & x < 0, \\
(\rho_r, u_r), & x > 0.
\end{cases}
\end{cases}
\tag{26}
$$

Solution. 第一个方程可变形为

$$\rho_t + \rho_x u + \rho u_x = 0.$$

第二个方程可变形为

$$\rho_t u + \rho u_t + (\rho u)_x u + (\rho u) u_x + p_x = 0.$$

即

$$\rho u_t + (\rho u) u_x + p_x = 0.$$

写成矩阵形式为

$$\begin{pmatrix} \rho \\ u \end{pmatrix}_t + \begin{pmatrix} u & \rho \\ \frac{p'(\rho)}{\rho} & u \end{pmatrix} \begin{pmatrix} \rho \\ u \end{pmatrix}_x = 0.$$

于是特征方程为

$$\begin{vmatrix} \lambda - u & \rho \\ \frac{p'(\rho)}{\rho} & \lambda - u \end{vmatrix} = 0.$$

解得两个互异特征值为

$$\lambda_1 = u - \sqrt{p'(\rho)}, \quad \lambda_2 = u + \sqrt{p'(\rho)}.$$

(i) *给定左状态 (ρ_ℓ, u_ℓ), 寻找合适的右状态 (ρ, u) 使得左右状态可用后向激波连接. 由 R-H 条件我们有*

$$\begin{cases} \sigma[\rho] = [\rho u], \\ \sigma[\rho u] = [\rho u^2 + p]. \end{cases}$$

即

$$\begin{cases} \sigma(\rho - \rho_\ell) = \rho u - \rho_\ell u_\ell, \\ \sigma(\rho u - \rho_\ell u_\ell) = \rho u^2 + p - (\rho_\ell u_\ell^2 + p_\ell). \end{cases}$$

由第一个关系式可得

$$\rho(u - \sigma) = \rho_\ell (u_\ell - \sigma).$$

由 *Lax* 熵条件

$$\lambda_1(r) < \sigma < \lambda_1(\ell)$$

可知 $\sigma < u_\ell$, 于是若设

$$m = \rho(u - \sigma) = \rho_\ell(u_\ell - \sigma),$$

则 $m > 0$. 因此

$$\sigma = u - \frac{m}{\rho} = u_\ell - \frac{m}{\rho_\ell}.$$

即

$$u - u_\ell = m\left(\frac{1}{\rho} - \frac{1}{\rho_\ell}\right).$$

从而 $u-u_\ell$ 与 $\rho-\rho_\ell$ 异号. 再由 *Lax* 熵条件

$$u-u_\ell<\sqrt{p'(\rho)}-\sqrt{p'(\rho_\ell)}$$

可知 $\rho-\rho_\ell>0$, 否则 $u-u_\ell$ 与 $\rho-\rho_\ell$ 同号. 于是 $u-u_\ell<0$. 又因为

$$\begin{aligned}\sigma(\rho u-\rho_\ell u_\ell)&=(\rho u-m)u-(\rho_\ell u_\ell-m)u_\ell\\&=\rho u^2-\rho_\ell u_\ell^2-m(u-u_\ell)=\rho u^2-\rho_\ell u_\ell^2+(p-p_\ell).\end{aligned}$$

故

$$-m(u-u_\ell)=p-p_\ell.$$

消去 m 可得

$$u-u_\ell=\pm\sqrt{\left(\frac{1}{\rho}-\frac{1}{\rho_\ell}\right)(p-p_\ell)}.$$

因为 $u<u_\ell$, 故

$$u=u_\ell-\sqrt{\left(\frac{1}{\rho}-\frac{1}{\rho_\ell}\right)(p-p_\ell)}\quad(\rho>\rho_\ell).$$

(ii) 给定右状态 (ρ_r,u_r), 寻找合适的左状态 (ρ,u) 使得左右状态可用前向激波连接. 由 *R-H* 条件我们有

$$\begin{cases}\sigma(\rho_r-\rho)=\rho_r u_r-\rho u,\\ \sigma(\rho_r u_r-\rho u)=\rho u_r^2+p_r-(\rho u^2+p).\end{cases}$$

与之前类似, 设

$$m=\rho(u-\sigma)=\rho_r(u_r-\sigma).$$

由 *Lax* 熵条件

$$\lambda_2(r)<\sigma<\lambda_2(\ell)$$

可知 $u_r<\sigma$, 从而 $m<0$. 又因为

$$\sigma=u-\frac{m}{\rho}=u_r-\frac{m}{\rho_r}.$$

故

$$u-u_r=m\left(\frac{1}{\rho}-\frac{1}{\rho_r}\right).$$

于是 $u-u_r$ 与 $\rho-\rho_r$ 同号. 再由 *Lax* 熵条件

$$u-u_r>\sqrt{p'(\rho_r)}-\sqrt{p'(\rho)}$$

可知 $u-u_r>0$, 否则 $u-u_r$ 和 $\rho-\rho_r$ 就会异号, 从而 $\rho-\rho_r>0$. 此时还可以确定 $\sigma>0$. 又因为

$$\sigma(\rho_r u_r-\rho u)=\rho_r u_r^2-\rho u^2-m(u_r-u)=\rho_r u_r^2-\rho u^2+p_r-p.$$

因此

$$-m(u_r-u)=p_r-p.$$

于是

$$u-u_r=\pm\sqrt{\left(\frac{1}{\rho}-\frac{1}{\rho_r}\right)(p_r-p)}.$$

又因为 $u>u_r$, 于是

$$u=u_r+\sqrt{\left(\frac{1}{\rho_r}-\frac{1}{\rho}\right)(p-p_r)}\quad(\rho>\rho_r).$$

(iii) 给定左状态 (ρ_ℓ, u_ℓ), 寻找合适的右状态 (ρ, u) 使得左右状态可用后向疏散波连接. 设 $\xi = x/t$, 则

$$\begin{pmatrix} u & \rho \\ \frac{p'(\rho)}{\rho} & u \end{pmatrix} \begin{pmatrix} \rho \\ u \end{pmatrix}_\xi = \xi \begin{pmatrix} \rho \\ u \end{pmatrix}_\xi .$$

此时 ξ 是矩阵

$$\begin{pmatrix} u & \rho \\ \frac{p'(\rho)}{\rho} & u \end{pmatrix}$$

的特征值, 相应的 $(\rho, u)^\top$ 为对应的右特征向量. 于是由第一个方程可知

$$\rho_\xi u + \rho u_\xi = \xi \rho_\xi \Rightarrow (\rho u)_\xi = \xi \rho_\xi .$$

即

$$\frac{\mathrm{d}(\rho u)}{\mathrm{d}\rho} = \xi = u - \sqrt{p'(\rho)}.$$

也即

$$u + \rho \frac{\mathrm{d}u}{\mathrm{d}\rho} = u - \sqrt{p'(\rho)} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}\rho} = -\frac{\sqrt{p'(\rho)}}{\rho}.$$

因为当 $\xi \uparrow$ 时, $\rho \downarrow$, 故 $\rho_\ell > \rho$ (如图 8). 于是

$$u = u_\ell - \int_{\rho_\ell}^{\rho} \frac{\sqrt{p'(y)}}{y} \mathrm{d}y \quad (\rho < \rho_\ell).$$

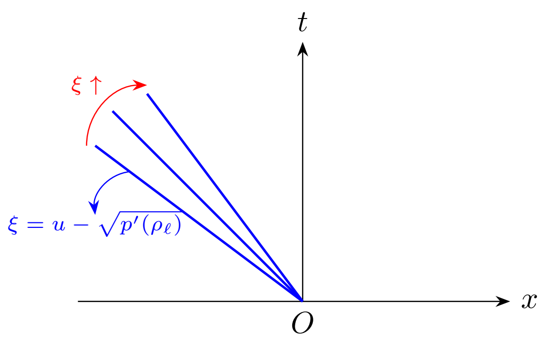

图 8: 后向疏散波线

(iv) 给定右状态 (ρ_r, u_r), 寻找合适的左状态 (ρ, u) 使得左右状态可用前向疏散波连接. 同样地, 我们有

$$\frac{\mathrm{d}u}{\mathrm{d}\rho} = \frac{\sqrt{p'(\rho)}}{\rho}.$$

当 $\xi \downarrow$ 时, $\rho \downarrow$, 故 $\rho > \rho_r$ (如图 9). 于是

$$u = u_r + \int_{\rho_r}^{\rho} \frac{\sqrt{p'(y)}}{y} \mathrm{d}y \quad (\rho > \rho_r).$$

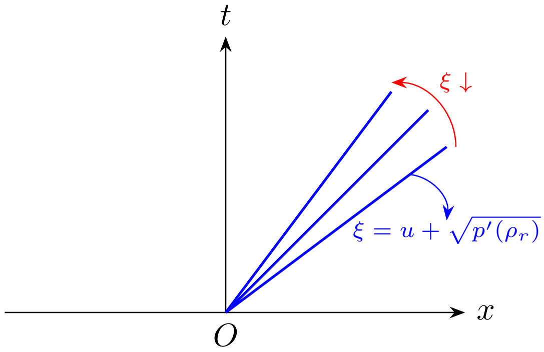

图 9: 前向疏散波线

$\square$

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